A Mini Project Report

On

Multimodal Identification Verication System Using Face and Speech Patterns

Submitted in Partial fulfillment of requirements for the award of the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

By

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P4: Demonstrate effective coding, written, presentation and oral communication skills.

MAPPING PROJECT OUTCOMES WITH PROGRAM OUTCOMES

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
P1	3	3	2	2	1	-	-	-	-	-	-
P2	2	3	3	2	1	1	1	-	-	-	2
P3	-	1	2	1	1	-	-	-	3	2	3
P4	-	-	-	-	1	-	-	1	2	3	1

1– LOW **2 –MEDIUM** **3– HIGH**

PROJECT OUTCOMES MAPPING WITH PROGRAM SPECIFIC OUTCOMES

PSO	PSO1	PSO2
P1	3	2
P2	3	2
P3	2	1
P4	1	1

PROJECT OUTCOMES MAPPING WITH PROGRAM EDUCATIONAL OBJECTIVES

PEO	PEO1	PEO2	PEO3	PEO4
P1	3	3	2	1
P2	2	3	2	1
P3	2	2	2	3
P4	2	1	2	3

DECLARATION

I hereby declare that the results embodied in the document entitled **“Multimodal Identification Verification System Using Face and Speech Patterns"** have been carried out by me during the academic year 2025-26 as a fulfillment for the completion of BE III year VI Semester Course Work of the B.E degree in Computer Science & Engineering from Osmania University. I have not submitted this document to any other university or organization for the award of any other degree.

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ABSTRACT

In this digital world, face recognition system is widely used in many areas such as security and identification. It is a type of biometric system that identifies a person using their facial features. Compared to other biometric systems like iris recognition or fingerprint authentication, face recognition is easier to use because it does not require physical contact. In many schools and colleges, attendance is taken manually by calling out student names, which takes up time and may also allow proxy attendance.To solve this problem, this project focuses on understanding and implementing a modern face recognition system based on deep learning. In this system, the faces are first recognised by using HOG method. This method makes the input image into black and white and converts it into HOG version of the image. The facial landmarks are recognised by using the dlib library. This is then sent to a previously trained Neural Network which generates 128 different and unique numerical vector values which identifies each face uniquely. This data is stored either in a CSV file or MySQL database.Using the Linear SVM classifier which is an ML classification algorithm, the differences between the measurements are understood and is used to recognise the person behind the specific measurements, and then a voice authentication system comes up proving a two factor authentication using MFCC model and SVM classifier to prevent giving proxy. This system helps in reducing manual work, saving time, and improving the accuracy of attendance management.

Keywords: Face Recognition, Deep Learning, Computer Vision, HOG (Histogram of Oriented Gradients), Facial Landmark Detection, MFCC, Whisper, Neural Networks, Attendance Management System, dlib Library, Machine Learning.

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CHAPTER-1

1.INTRODUCTION

1.1 PURPOSE OF THE PROJECT

The purpose of this project is to develop a secure and automated attendance management system using face recognition and voice authentication technologies. Traditional attendance methods are time-consuming, require manual effort, and are vulnerable to proxy attendance. To overcome these limitations, the proposed system uses biometric verification to accurately identify students and record their attendance.

The system employs face recognition based on facial embeddings generated by a pre-trained deep learning model and verifies the identity of the student using Euclidean distance matching. To further enhance security and prevent impersonation, a voice authentication module is integrated using MFCC feature extraction and an SVM classifier. Attendance is marked only when the student's roll number, facial features, and voice characteristics match the stored records.

The project aims to reduce manual workload, improve attendance accuracy, prevent unauthorized attendance marking, and demonstrate the practical application of biometric authentication techniques in educational institutions. The attendance records are automatically maintained in a CSV file, making the system simple, efficient, and easy to manage.

1.2 SCOPE OF THE PROJECT

The scope of this project is to design and implement a biometric-based attendance management system that uses both face recognition and voice authentication for student verification. The system allows students to register by providing their roll number, facial images, and voice samples, which are processed and stored as biometric templates in a custom dataset.

During attendance marking, the student enters their roll number and provides live face and voice samples. The system extracts facial embeddings and voice features, compares them with the stored templates using similarity metrics, and grants attendance only when all verification conditions are satisfied. The attendance status is then recorded automatically in a CSV file.

The project demonstrates the practical application of computer vision, machine learning, and biometric authentication techniques in educational institutions. It reduces manual

effort, minimizes proxy attendance, and improves the reliability of attendance records. Although the current implementation is designed for classroom attendance management, the same approach can be extended to offices, organizations, examination centers, and access-control systems.

Future enhancements may include real-time attendance monitoring, cloud-based database integration, mobile application support, anti-spoofing mechanisms, and deployment on large-scale institutional networks.

1.3 PROBLEMS WITH EXISTING SYSTEMS

Traditional attendance management systems used in schools and colleges are largely manual, requiring faculty members to call out student names and record attendance. This process consumes valuable classroom time and is prone to human errors such as incorrect entries or missed records. Moreover, manual systems cannot effectively prevent proxy attendance, where one student answers on behalf of another.

Many existing automated attendance systems rely on a single biometric factor, such as face recognition or fingerprint authentication. Single-biometric systems are vulnerable to spoofing attacks, misidentification, and reduced accuracy under varying environmental conditions. Face recognition systems alone may be affected by changes in lighting, facial expressions, or image quality, while voice-based systems may be influenced by background noise and voice variations.

Hardware-dependent biometric systems often require specialized devices such as fingerprint scanners or iris sensors, increasing deployment and maintenance costs. Additionally, some existing systems lack a reliable mechanism for verifying user identity through multiple authentication factors.

Therefore, there is a need for a secure, accurate, and cost-effective attendance management system that minimizes proxy attendance, reduces manual effort, and provides stronger identity verification through the combination of face recognition and voice authentication.

1.4 ARCHITECTURE DIAGRAM

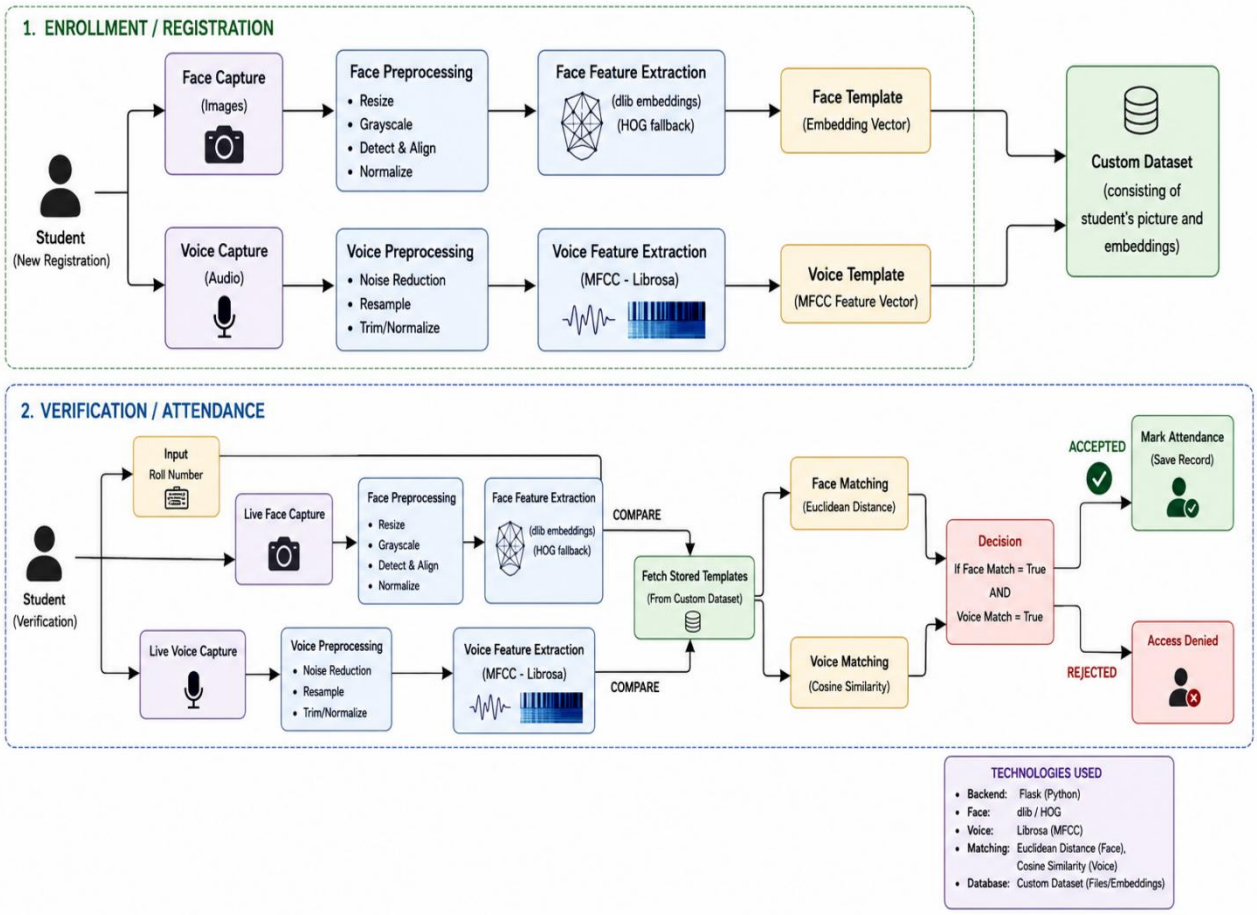


Fig 1.4 Face and Voice Attendance System Architecture

The system uses a multi-modal biometric approach that combines face recognition and voice authentication for attendance management. During registration, facial and voice features are extracted and stored as biometric templates in a custom datasets. During verification, live face and voice samples are matched with the stored templates using similarity measures. Attendance is marked only when both face and voice authentication are successfully verified.

CHAPTER-2

2.LITERATURE SURVEY

2.1 OVERVIEW

In the recent years, deep learning and computer vision techniques have significantly improved in the field of facial biometric and vocal biometric recognition. Researchers have developed various deep learning and machine learning to identify features from facial images and audio files. earlier approaches relied on traditional and basic techniques, while modern approaches utilize Convolution Neural Networks (CNN) and Deep Learning for higher accuracy. This chapter presents a review of existing facial detection methodologies, their limitations and the motivation behind selecting Haar Cascade, dlib, MFCC and SVM classifier.

2.2 REVIEW OF LITERATURE

Dalal and Triggs [1] introduced the **Histogram of Oriented Gradients (HOG)** algorithm for object and face detection. The HOG descriptor extracts gradient orientation features from facial images, providing robust detection under varying illumination conditions. Due to its computational efficiency and reliable performance, HOG has become one of the most widely adopted feature extraction techniques in face recognition systems.

King [2] developed the **dlib Machine Learning Toolkit**, which provides efficient facial landmark detection and face recognition algorithms. The library identifies important facial landmarks such as eyes, nose, and mouth, enabling accurate facial alignment before feature extraction. Dlib is extensively used in modern face recognition applications because of its high detection accuracy and ease of implementation.

Schroff et al. [3] proposed **FaceNet**, a deep learning framework that generates a unique 128-dimensional embedding vector for every human face. Instead of directly classifying facial images, FaceNet learns an embedding representation where similar faces are positioned close together in feature space. This approach significantly improves face verification and identification accuracy.

Viola and Jones [4] introduced the **Haar Cascade** algorithm for rapid object detection. The algorithm employs Haar-like features and the AdaBoost classifier to detect faces

efficiently in real time. Although newer deep learning methods provide higher accuracy, Haar Cascade remains popular because of its simplicity and fast processing speed.

Turk and Pentland [5] proposed the **Eigenfaces** method for face recognition using Principal Component Analysis (PCA). The algorithm represents faces as weighted combinations of eigenvectors, reducing computational complexity while maintaining satisfactory recognition performance. This work laid the foundation for many modern face recognition techniques.

Davis and Mermelstein [6] introduced **Mel Frequency Cepstral Coefficients (MFCC)** for speech feature extraction. MFCC captures important frequency characteristics of human speech by converting audio signals into compact numerical representations. It remains one of the most widely used techniques for speaker recognition and voice authentication.

Reynolds [7] proposed the use of **Gaussian Mixture Models (GMM)** for speaker identification and verification. GMM effectively models the statistical distribution of voice features extracted from speech signals and improves speaker recognition under different recording environments.

Cortes and Vapnik [8] introduced the **Support Vector Machine (SVM)** algorithm for classification tasks. SVM constructs an optimal decision boundary that separates different classes with maximum margin, resulting in high classification accuracy. It has become a widely used classifier in biometric authentication systems, particularly for voice verification.

Radford et al. [9] developed **Whisper**, a deep learning speech recognition model trained on large-scale multilingual datasets. Whisper demonstrates excellent robustness against background noise and different accents, making it suitable for real-world voice processing applications.

Jain, Ross, and Prabhakar [10] presented a comprehensive survey on **multimodal biometric recognition**, demonstrating that combining multiple biometric traits such as face and voice provides significantly higher authentication accuracy than using a single

biometric modality. Their research showed that multimodal systems greatly reduce spoofing attacks and improve system reliability.

From the above literature, it is observed that face recognition techniques such as HOG, dlib, FaceNet, and Eigenfaces provide reliable facial identification, while MFCC, GMM, SVM, and Whisper improve speaker verification performance. However, many existing systems rely on a single biometric trait, making them more vulnerable to spoofing and impersonation attacks. Therefore, integrating both face recognition and voice authentication provides a more secure, reliable, and accurate biometric attendance management system suitable for educational institutions.

2.3 SUMMARY AND RESEARCH GAPS

2.3.1 Research Gap Table

Table 2.3.1 Research Table

Authors	Proposed Model	Features	Findings	Limitations
Dalal & Triggs (2005)	HOG	Gradient-based face detection	Fast face detection	Sensitive to pose variation
King (2009)	dlib	Facial landmark detection	High face recognition accuracy	Performance depends on image quality
Schroff et al. (2015)	FaceNet	128-D Face Embeddings	Excellent recognition accuracy	Requires large datasets
Davis & Mermelstein (1980)	MFCC	Speech feature extraction	Effective voice representation	Sensitive to background noise
Reynolds (1995)	GMM	Speaker verification	Reliable voice modeling	Lower performance with limited data
Cortes & Vapnik (1995)	SVM	Classification algorithm	High classification accuracy	Depends on feature quality
Radford et al. (2022)	Whisper	Speech recognition	Robust multilingual recognition	High computational requirements
Jain et al. (2004)	Multimodal Biometrics	Face & Voice Fusion	Improved authentication security	Complex implementation
Zhang et al. (2016)	MTCNN	Face detection & alignment	Accurate facial localization	Computationally intensive
Goodfellow et al. (2016)	Deep Learning	Neural network models	High recognition accuracy	Requires powerful hardware

2.3.2 Research Gaps

Although recent studies have achieved significant improvements in face recognition and voice authentication, several challenges still exist in developing a highly secure and reliable biometric attendance system. Existing approaches mainly focus on individual biometric modalities, resulting in limitations related to security, scalability, computational efficiency, and real-world deployment. Addressing these research gaps is essential for building an accurate and robust multimodal identity verification system.

- **Lack of Multi-Modal Authentication**

Most existing attendance systems rely on either face recognition or voice authentication independently. Using a single biometric modality increases the possibility of spoofing, impersonation, and authentication failure under varying environmental conditions. A multimodal biometric system combining both face and voice verification can significantly improve authentication accuracy and system reliability.

- **Sensitivity to Environmental Conditions**

Face recognition performance decreases under poor lighting conditions, different facial poses, occlusions, and image quality variations. Similarly, voice authentication accuracy is affected by background noise, microphone quality, and speech variations. These environmental factors reduce the effectiveness of existing biometric systems in real-world scenarios.

- **Limited Scalability and Data Management**

Many attendance systems are designed for small datasets and are unable to efficiently manage large numbers of users. The absence of cloud-based storage and centralized database management limits system scalability and makes deployment difficult for large educational institutions and organizations.

- **Security and Spoofing Attacks**

Several face recognition systems are vulnerable to spoofing attacks using photographs or recorded videos, while voice authentication systems can be deceived through replay attacks. Existing solutions provide limited protection against these security threats, making stronger anti-spoofing mechanisms necessary.

- **High Computational Complexity**

Advanced deep learning models require powerful hardware, large datasets, and significant computational resources for training and deployment. This increases implementation cost and makes real-time execution difficult on low-resource systems.

Summary

The literature survey shows that significant progress has been achieved in face recognition and speaker verification using machine learning and deep learning techniques. However, existing systems still suffer from challenges such as environmental sensitivity, spoofing attacks, limited scalability, and dependence on single biometric authentication. These limitations motivate the development of the proposed Multimodal Identification Verification System, which combines facial recognition and voice authentication to improve security, reliability, and attendance accuracy.

2.3.3 Comparison of Existing Systems

The following table presents a comparison of the existing face recognition and voice authentication techniques discussed in the literature. The comparison is based on the technology used, application area, reported accuracy, and major limitations. This analysis highlights the need for a multimodal biometric system that combines face recognition and voice authentication to achieve higher security and accuracy.

Table 2.3.3 Comparison of Existing Systems

Ref. No	Method	Application	Accuracy	Limitations
[1]	HOG	Face Detection	89%	Sensitive to pose and lighting changes
[2]	dlib	Facial Landmark Detection	92%	Image quality affects performance
[3]	FaceNet	Face Recognition	99.63%	Requires large training datasets
[4]	Haar Cascade	Face Detection	90%	Less accurate under complex backgrounds
[5]	Eigenfaces	Face Recognition	91%	Affected by illumination variations
[6]	MFCC	Voice Feature Extraction	91%	Sensitive to background noise
[7]	GMM	Speaker Verification	90%	Lower performance with limited samples
[8]	SVM	Voice Classification	94%	Depends on feature extraction quality
[9]	Whisper	Speech Recognition	95%	Computationally intensive
[10]	Multimodal Biometrics	Face + Voice Authentication	97%	Higher implementation complexity

2.3.4 Accuracy Analysis

The comparison of existing methods indicates that deep learning-based face recognition models provide higher accuracy than traditional approaches. FaceNet achieves one of the highest recognition accuracies, while MFCC combined with SVM provides reliable speaker verification. Multimodal biometric systems that combine face recognition and voice authentication offer better security and improved overall performance compared to systems using only a single biometric modality.

Table 2.3.4 Accuracy Comparison

Technique	Accuracy (%)
HOG	89
dlib	92
FaceNet	99.63
Haar Cascade	90
Eigenfaces	91
MFCC	91
GMM	90
SVM	94
Whisper	95
Multimodal Biometrics	97

CHAPTER-3

3.SOFTWARE REQUIREMENT SPECIFICATION

3.1 INTRODUCTION TO SRS

Software Requirement Specification (SRS) is a detailed description of the functional and non-functional requirements of a software system. It serves as a blueprint for the design, development, implementation, and testing phases of the project.

The proposed system, "Multimodal Identification Verification System Using Face and Speech Patterns", is developed to automate attendance management using biometric authentication. The system utilizes facial recognition and voice authentication to verify the identity of students before marking attendance. Students are required to register using their roll number, facial images, and voice samples. During attendance marking, the system compares the live face and voice data with the stored biometric templates. Attendance is recorded only when both verifications are successful. The attendance records are automatically stored in a CSV file for future reference and management

3.2 FUNCTIONAL REQUIREMENTS

The functional requirements describe the services and operations that the system must perform.

Table 3.2 Functional Requirements

Module	Functional Requirements
Student Registration Module	Register students using roll number
Student Registration Module	Capture facial images using webcam
Student Registration Module	Record voice samples using microphone
Student Registration Module	Generate and store facial embeddings and voice templates
Face Recognition Module	Detect faces using Haar Cascade or HOG
Face Recognition Module	Identify facial landmarks using dlib
Face Recognition Module	Generate 128-dimensional facial embeddings
Face Recognition Module	Compare embeddings using Euclidean Distance
Voice Authentication Module	Record voice input from student
Voice Authentication Module	Extract voice features using MFCC
Voice Authentication Module	Verify voice using SVM classifier
Voice Authentication Module	Compare voice features using Cosine Similarity
Attendance Management Module	Verify roll number, face, and voice
Attendance Management Module	Mark attendance after successful verification
Attendance Management Module	Prevent unauthorized attendance marking
Attendance Management Module	Store attendance records automatically in CSV
Data Storage Module	Maintain biometric templates
Data Storage Module	Store attendance logs locally

3.3 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes and performance standards that the system must satisfy for efficient and reliable operation.

Table 3.3 Non Functional Requirements

Category	Requirement
Performance	Face and voice verification should complete within a few seconds
Performance	Attendance marking should occur in real time
Reliability	Provide accurate face and voice verification
Reliability	Maintain attendance records without data loss
Security	Attendance only after successful biometric verification
Security	Minimize proxy attendance
Security	Secure storage of biometric templates
Usability	User interface should be simple and easy to use
Usability	Students should register and mark attendance with minimal effort
Maintainability	Easy updating of student records
Maintainability	Support integration of future biometric modules
Scalability	Support registration of multiple students
Scalability	Extendable to larger educational institutions

3.4 SOFTWARE REQUIREMENTS

Software requirements specify the software tools, programming languages, libraries, and platforms needed to develop and run the system.

Table 3.4 Software Requirements

Software	Description
Windows 10/11	Operating System
Python 3.x	Programming Language
VS Code	Development Environment
OpenCV	Image Processing
dlib	Face Detection
Flask	Backend Framework
Librosa	Audio Processing
Scikit-learn	Machine Learning
GitHub	Version Control

3.5 HARDWARE REQUIREMENTS

Hardware requirements specify the physical devices and system resources needed for the development and deployment of the system.

Table 3.5 Hardware Requirements

Hardware	Specification
Processor	Intel Core i5+
RAM	8 GB Minimum
Storage	10 GB Free Space
Webcam	Face Capture
Microphone	Voice Recording
Monitor	1366 × 768 Resolution
Keyboard	Input Device
Mouse	Input Device

CHAPTER-4

4.SYSTEM DESIGN

4.1 INTRODUCTION TO UML

Unified Modeling Language (UML) is a standardized visual modeling language used to analyze, design, and document software systems. It provides a set of graphical notations that help developers represent the structure, behavior, and interactions of various components within a system. UML acts as a bridge between system requirements and implementation by providing a clear visualization of how the system operates.

In the proposed project, "Multimodal Identification Verification System Using Face and Speech Patterns", UML is used to model the overall architecture and workflow of the attendance management system. It helps in illustrating how students interact with the system during registration and attendance marking processes. UML diagrams provide a clear understanding of the relationships between different modules such as face detection, face recognition, voice authentication, attendance management, and data storage.

Various UML diagrams such as Use Case Diagrams, Activity Diagrams, Sequence Diagrams, Class Diagrams, and Data Flow Diagrams can be used to represent the system from different perspectives. These diagrams simplify system analysis, improve communication among developers, and assist in identifying system requirements before implementation.

By using UML, the proposed system can be designed in a structured manner, ensuring better maintainability, scalability, and overall software quality.

4.2 UML DIAGRAMS

4.2.1 USE CASE DIAGRAM

A behavioral diagram that shows the interaction between the student and the attendance management system. It represents the different functions available to the user such as student registration, face capture, voice recording, biometric verification, and attendance

marking.

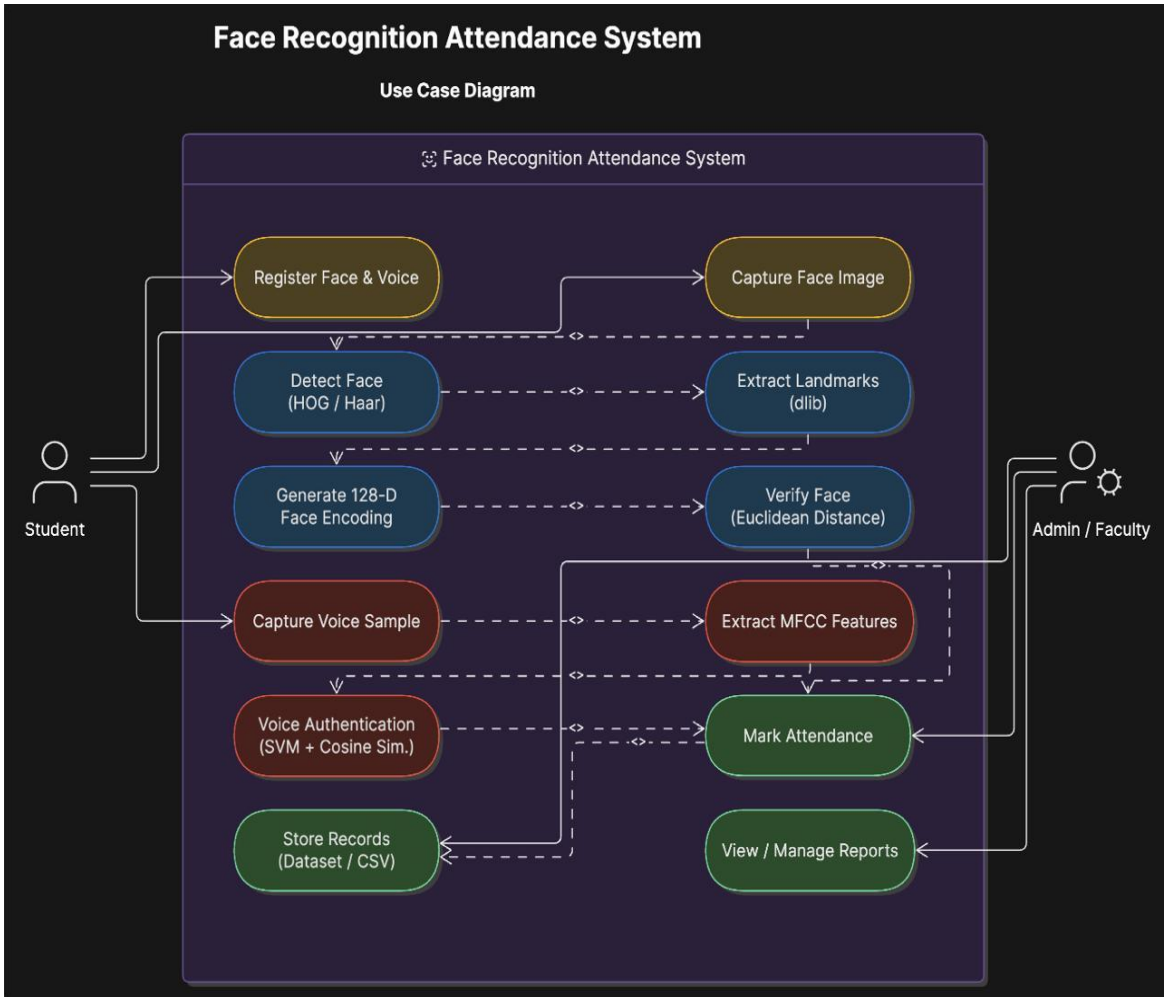


Fig 4.2.1 UseCase Diagram

The Use Case Diagram illustrates the interactions between the student and the attendance system. It shows the main functionalities such as student registration, face capture, voice recording, biometric verification, and attendance marking. The diagram represents how users interact with different modules of the system to complete attendance-related activities.

4.2.2 CLASS DIAGRAM

A structural diagram that represents the static structure of the system. It shows the main classes used in the project such as Student, Face Recognition Module, Voice Authentication Module, Attendance Manager, and Database along with their attributes and relationships.

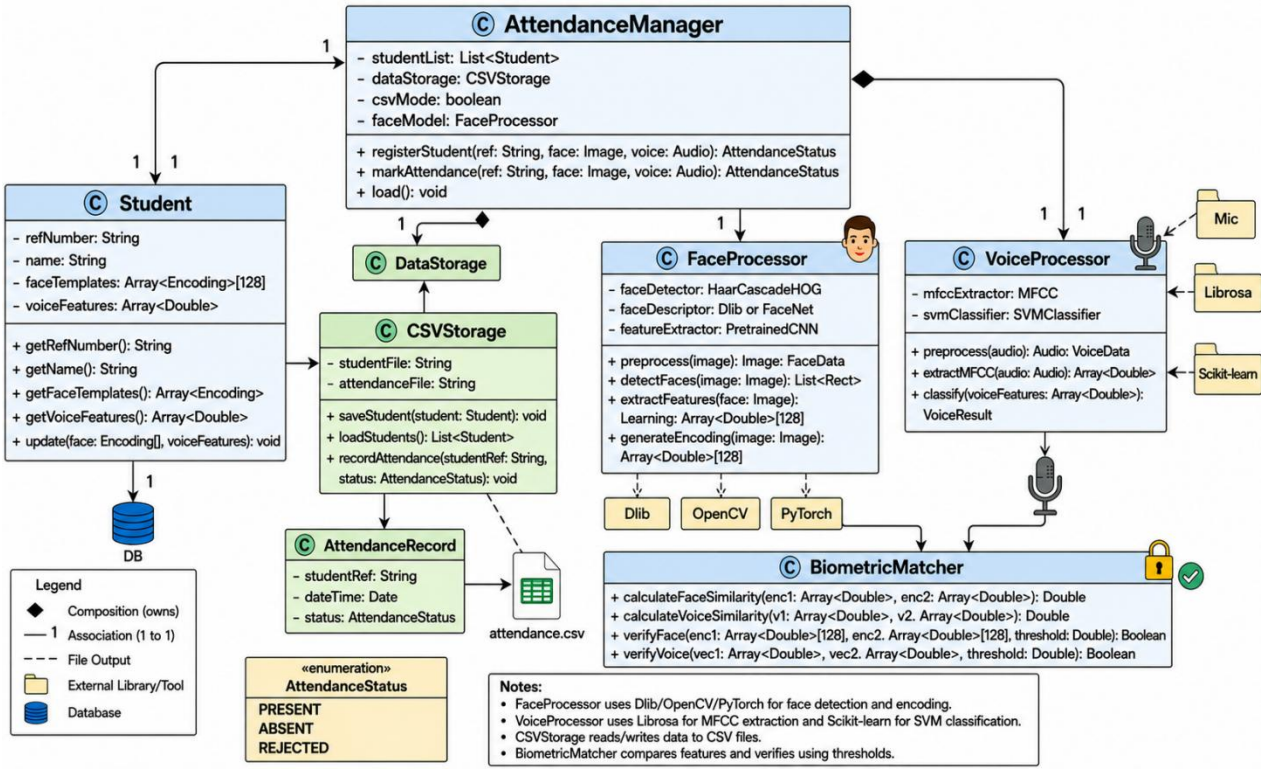


Fig 4.2.2 Class Diagram

The Class Diagram represents the static structure of the system and the relationships between its classes. It includes classes such as Student, Face Recognition Module, Voice Authentication Module, Attendance Manager, and Database. The diagram shows the attributes and methods used for storing biometric data and managing attendance records.

4.2.3 SEQUENCE DIAGRAM

A behavioral diagram that shows how objects interact in a time-ordered sequence. It focuses on the exchange of messages between components of the system.

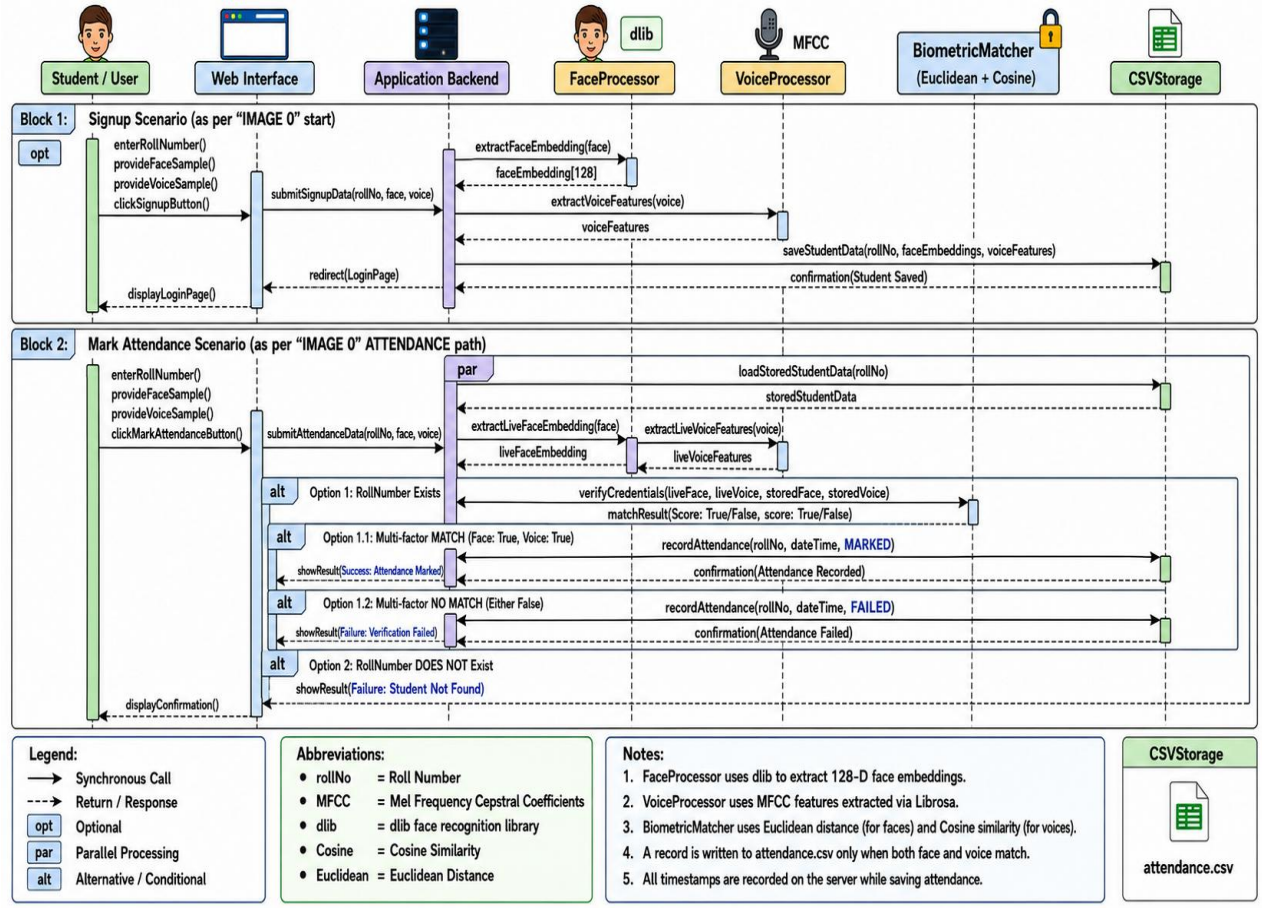


Fig 4.2.3 Sequence Diagram

The Sequence Diagram illustrates the flow of messages exchanged between different components during registration and attendance verification. It shows how the student, system, face recognition module, voice authentication module, and database interact in a time-ordered sequence. The diagram helps visualize the step-by-step authentication process.

4.2.4 ACTIVITY DIAGRAM

An Activity Diagram shows the flow of activities in a system from start to end. It represents the sequence of steps involved in student registration, verification, and attendance marking.

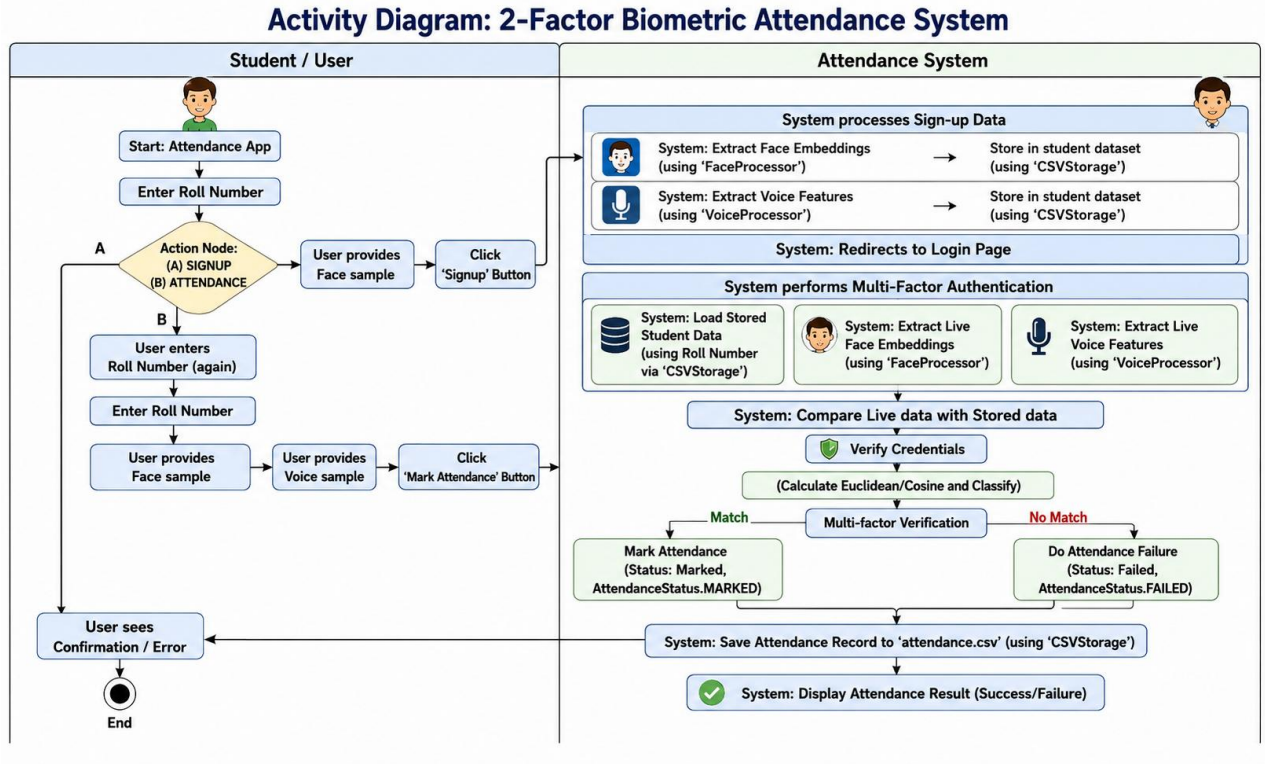


Fig 4.2.4 Activity Diagram

The Activity Diagram shows the workflow of the attendance management system from registration to attendance marking. It represents activities such as face capture, voice recording, biometric verification, decision making, and attendance storage. The diagram clearly illustrates the sequence of actions performed by the system.

4.2.5 STATE CHART DIAGRAM

A behavioral diagram that describes the different states of an object/system and how it changes from one state to another based on events.

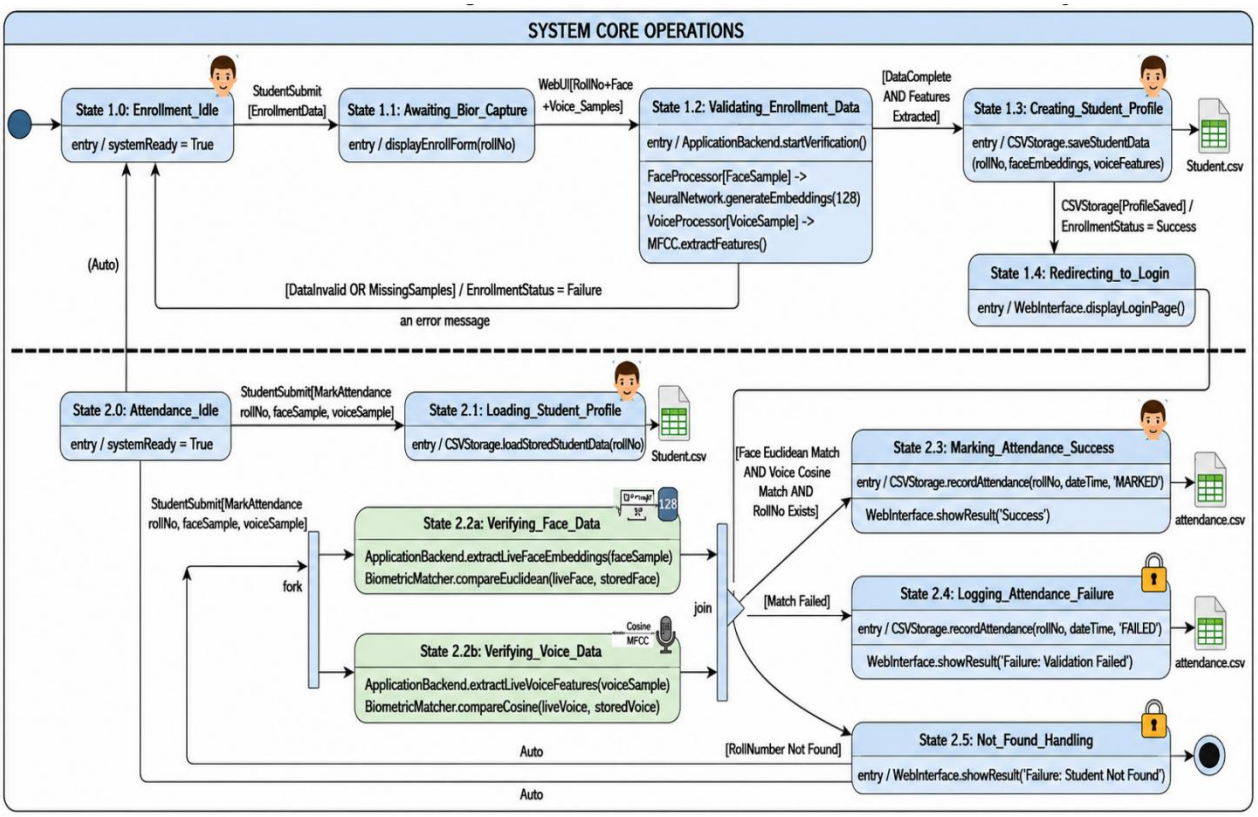


Fig 4.2.5 Statechart Diagram

The State Chart Diagram describes the various states of the attendance system and transitions between them. The system moves through states such as registration, waiting for input, verification, attendance marked, and authentication failed. It helps in understanding the behavior of the system during execution.

4.2.6 DEPLOYMENT DIAGRAM

A structural diagram that shows the physical deployment of the system. It represents how software components are distributed across hardware nodes like client, server, and database.

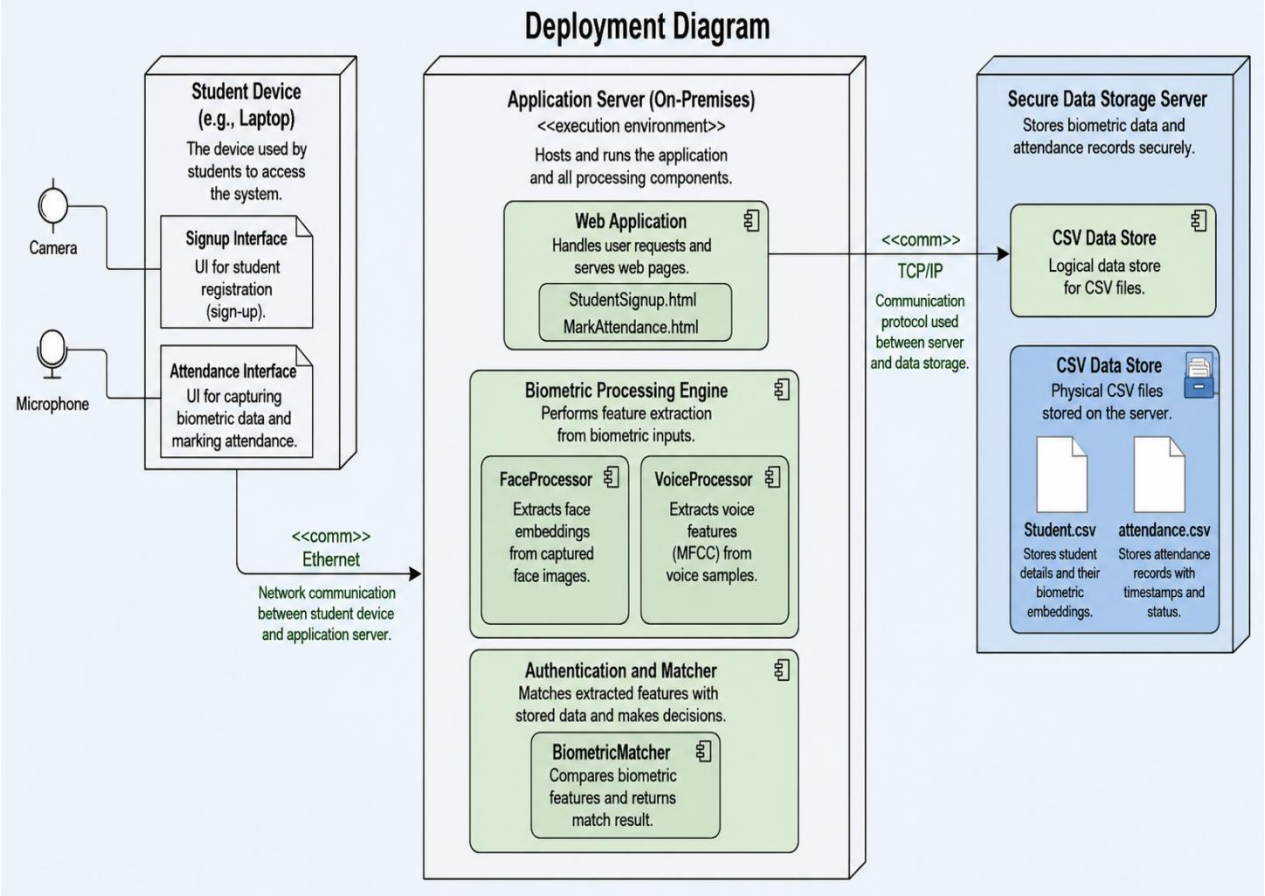


Fig 4.2.6 Deployment Diagram

The Deployment Diagram shows how the software components are deployed on the hardware infrastructure. It represents the interaction between the client device, webcam, microphone, application server, and data storage system. The diagram provides a clear view of the physical architecture of the attendance management system.

4.2.7 FRONTEND WORKFLOW

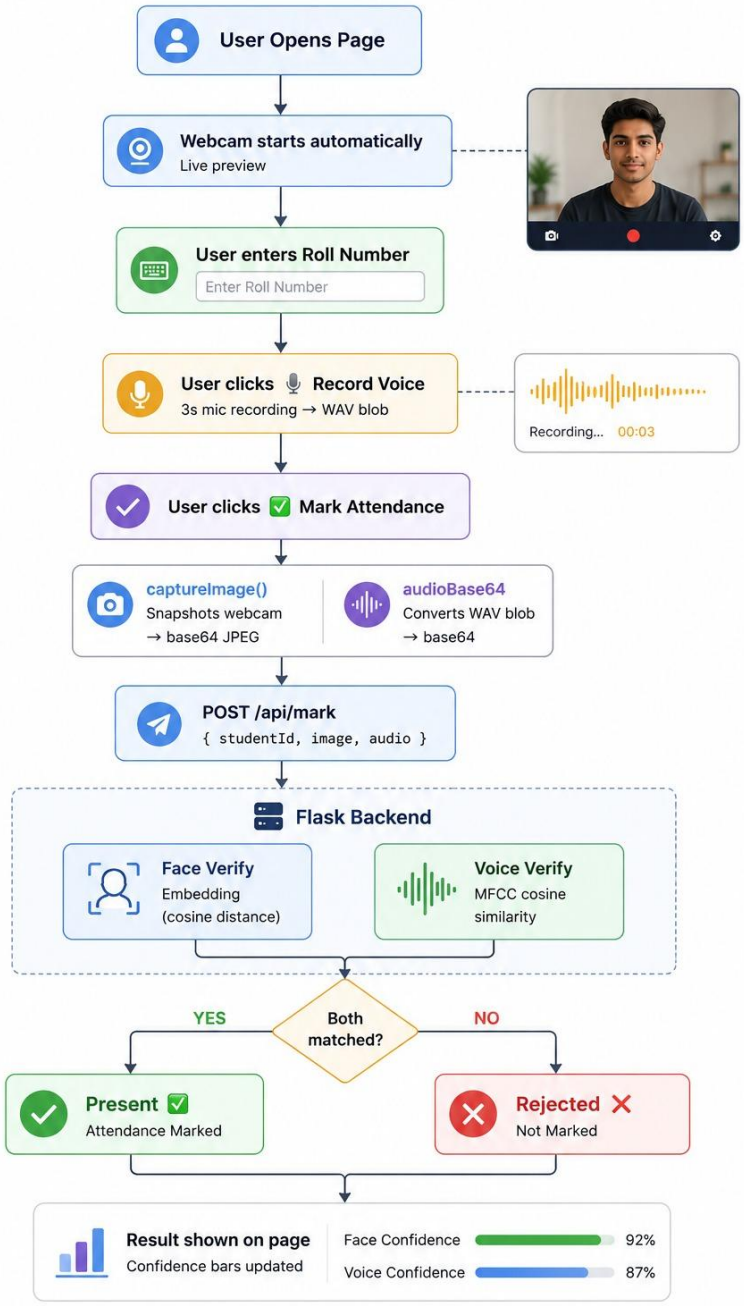


Fig 4.2.7 Frontend Workflow

The Frontend Workflow illustrates the sequence of user interactions with the web interface. It includes student registration, face and voice capture, verification requests, authentication results, and attendance confirmation. The workflow helps explain how users navigate through the system from start to finish.

CHAPTER-5

5.IMPLEMENTATION

5.1 TECHNOLOGIES USED

The proposed attendance management system is developed using a combination of web technologies, machine learning libraries, and deep learning models. The frontend is built using HTML5, CSS3, and Vanilla JavaScript to provide an interactive user interface, while the backend is implemented using Python and the Flask framework. Face recognition is performed using dlib, OpenCV, and HOG-based image processing techniques, whereas voice authentication is implemented using Librosa for audio feature extraction.

The system uses Euclidean Distance for face matching and Cosine Similarity for voice matching. Student information and attendance records are stored using JSON, CSV, and NumPy (.npy) file formats. Version control is maintained using Git and GitHub. For deep learning, MobileNetV2 is used to generate facial embeddings, and an LSTM Speaker Encoder is employed to generate voice embeddings, both implemented using the PyTorch framework.

Table 5.1 Technologies Used

Layer / Component	Technology Used
Frontend	HTML5, CSS3, Vanilla JavaScript
Backend	Python, Flask
Machine Learning Service	face-recognition (dlib), Librosa, scikit-image (HOG), OpenCV
Similarity Measure (Face & Voice)	Euclidean Distance (Face), Cosine Similarity (Voice)
Data Format	CSV (Attendance), JSON (Student Registry), .npy (Embeddings)
Version Control	Git, GitHub
Deep Learning Model	MobileNetV2 (Face Embeddings), LSTM Speaker Encoder (Voice D-Vectors), PyTorch

5.2 METHODOLOGY

The implementation of the **Multimodal Identification Verification System Using Face and Speech Patterns** follows a systematic methodology to provide secure, accurate, and automated attendance management using two-factor biometric authentication. The process begins with the student enrollment phase, where each student registers by entering their roll number, capturing facial images through a webcam, and recording a voice sample using a microphone. The captured face images are preprocessed using OpenCV and dlib to detect facial landmarks and generate a unique 128-dimensional facial embedding using a pre-trained deep learning model. Similarly, the voice sample is processed using the Librosa library to extract Mel Frequency Cepstral Coefficients (MFCC), which represent the unique characteristics of the student's voice. These face embeddings and voice features are securely stored along with the student's roll number in local storage files such as CSV, JSON, and NumPy formats. During the attendance verification phase, the student provides the registered roll number along with live face and voice samples. The system extracts biometric features from the live inputs and compares them with the stored templates using Euclidean Distance for face recognition and Cosine Similarity for voice authentication. Attendance is marked only when both biometric verifications satisfy the predefined threshold values, ensuring a reliable two-factor authentication mechanism that prevents proxy attendance and unauthorized access.

The entire application is implemented using **Python** and the **Flask** framework for backend processing, while **HTML5, CSS3, and JavaScript** are used to develop the interactive frontend interface. OpenCV, dlib, Librosa, PyTorch, and Scikit-learn are integrated to perform computer vision, speech processing, feature extraction, and machine learning tasks efficiently. Once authentication is successful, the system automatically records the student's attendance along with the date and time in a CSV file and displays the verification result on the web interface. This implementation methodology ensures accurate identity verification, minimizes manual effort, enhances security, and provides an efficient, scalable, and user-friendly attendance management solution suitable for educational institutions.

5.3 SAMPLE CODE

5.3.1 Deep Learning Model Implementation

```
#d1_models. py
import torch, torch.nn as nn, torch.nn. functional as F from
torchvision import models
class FaceEmbeddingNet(nn. Module) :
"'"MobileNetV2 backbone → 128D face embedding"''"
def _init_:
super ().__init__()
base = models.mobilenet_v2(weights="DEFAULT")
self. features = base. features
self.pool= nn.AdaptiveAvgPoo12d((1, 1))
self.fc = nn. Linear (1280, 128)
def forward (self, x):
x= self. pool self. features (x))
return F.normalize(self.fc(torch. flatten (x, 1)), dim=1)
class SpeakerEncoderLSTM(nn. Module) :
"''"3-layer LSTM → 256D speaker (voice) embedding"'"' def
_init_:
super ().__init__()
self.Istm = nn. LSTM(40, 256, num_layers=3, batch_first=True)
self.fc = nn. Linear(256, 256)
def forward (self, x):
out, _ = self. 1stm(x)
return F.normalize(self.fc(outl:, -1, :l), dim=1)
def cosine_verify(stored, query, threshold=0.75):
''''Compare query embedding against stored embeddings."''"
scores = stored @query / (torch.norm(stored, dim=1) *
torch.norm(query) + 1e-9)
best = scores. max ().item)
return {"matched": best >= threshold, "score": round(best,
4)}
```

5.3.2 Backend Implementation Using Flask

```
#web_app•py
from flask import Flask, request, jsonify from d1_models
import FaceEmbeddingNet, SpeakerEncoderLST, cosine_verify
import torch, base64 from pathlib import Path
app = Flask(name )
face_model = FaceEmbeddingNet). eval)
voice_model = SpeakerEncoderLSTM() . eval)
store = 0
# (student_id: {"face": tensor, "voice": tensor}}
@app.post("/api/enroll")
def enroll():
data = request. get_json()
sid = data[ "studentId"]
store[sid] = {
"face":
extract_face(data["image"1),
"voice":
extract_voice(data[ "audio"]),
}
return jsonify({"ok": True})
@app.post("/api/verify")
def verify():
data = request.get_json()
sid = data[ "studentId"]
if sid not in store:
return jsonify(f"ok": False, "error": "Not enrolled"}), 404
face_r = cosine_verify(store[sid]l"face"],
extract_face(data["image"]))
voice_r = cosine_verify(store[sid]["voice"],
extract_voice(data["audio"]))
return jsonify({
"accepted": face_r[ "matched"1 and voice_r[ "matched"],
"face": face r, "voice": voice r
```

5.3.3 Frontend Implementation

HTML

```
<!-- Webcam live feed -->
<video id="camera" autoplay muted></video>

<!-- Roll number input -->
<input id="rollNo" placeholder="Enter Roll Number" />

<!-- Voice record toggle -->
<button id="recordBtn">🎙 Record Voice</button>

<!-- Submit -->
<button id="markBtn">✅ Mark Attendance</button>

<!-- Result banner -->
<div id="result"></div>
```

JavaScript

```
// 1. Start webcam
navigator.mediaDevices.getUserMedia({ video: true })
  .then(stream => camera.srcObject = stream);
// 2. Capture face from webcam → base64
function captureImage() {
  const canvas = document.createElement("canvas");
  canvas.getContext("2d").drawImage(camera, 0, 0);
  return canvas.toDataURL("image/jpeg");  // base64 string
}
// 3. Record mic → WAV blob → base64
recordBtn.onclick = async () => {
  const stream = await
navigator.mediaDevices.getUserMedia({ audio: true });
  const recorder = new MediaRecorder(stream);
  recorder.start();
```

```
  setTimeout(() => recorder.stop(), 3000);  // 3 sec
recording
};

// 4. Send to Flask backend
markBtn.onclick = async () => {
  const res = await fetch("/api/mark", {
    method: "POST",
    body: JSON.stringify({
      studentId: rollNo.value,
      image: captureImage(),   // face snapshot
      audio: audioBase64,      // voice recording
    })
  });
  const data = await res.json();
  result.textContent = data.accepted ? "✅ Present" : "❌
Rejected";
};
```

5.4 CONFIDENCE ANALYSIS

The system evaluates biometric identity through two independent confidence scores — one for face verification and one for voice verification. For face verification, confidence is computed from the Euclidean distance between the stored 128-dimensional enrollment embedding and the live captured embedding, where a lower distance indicates a stronger match. This distance is converted into a confidence percentage using the formula confidence = 1 − (distance / 1.2), clamped between 0 and 1. A match is accepted when the distance falls below a threshold of 0.60 using the dlib face recognition method, or when cosine similarity exceeds 0.85 under the HOG fallback method. For voice verification, confidence is measured using Cosine Similarity between the stored MFCC template and the live recorded voice features, where a score closer to 1.0 indicates a near-identical voice pattern. The system compares the live query against all stored templates and selects the highest score, accepting a match when the score exceeds 0.40 for standard 13-dimensional MFCC vectors. Both modalities must independently pass their respective

thresholds for the student to be marked as present — if either the face or voice verification fails, the attendance is rejected. The resulting confidence scores are displayed in real time on the login interface as percentage progress bars, providing immediate visual feedback. This dual-threshold, dual-modality design ensures that neither biometric alone is sufficient, significantly reducing the risk of impersonation or spoofing.

5.5 SUMMARY

This chapter explained the implementation of the Multimodal Identification Verification System using face and speech patterns. The system was developed using Python, OpenCV, dlib, HOG, Face Recognition, MFCC, and SVM algorithms for biometric authentication.

The implementation includes student registration, face detection, facial feature extraction, voice feature extraction, identity verification, and attendance marking. Face recognition and voice authentication work together as a two-factor authentication mechanism to prevent proxy attendance and improve security.

The system stores biometric data and attendance records efficiently, providing accurate and automated attendance management. The successful integration of computer vision, machine learning, and speech processing technologies demonstrates the effectiveness of the proposed solution.

CHAPTER-6

6.RESULTS

6.1 OUTPUT SCREENS

6.1.1 Student Registration Interface

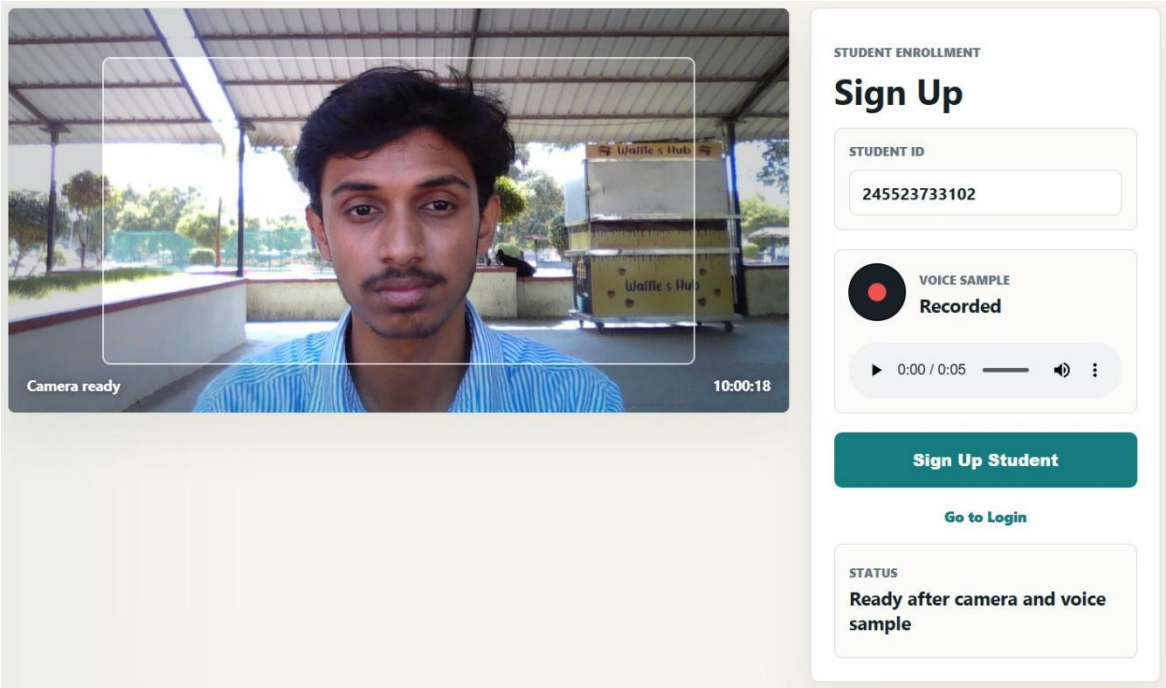


Fig 6.1.1 Student Registration Interface

The registration interface allows a student to enroll into the system by entering the roll number, capturing a facial image through the webcam, and recording a voice sample through the microphone. The captured face image is processed to generate facial embeddings, while the voice sample is converted into MFCC features and stored in the dataset. This registration process creates the biometric templates required for future identity verification.

6.1.2 Student Login and Biometric Verification

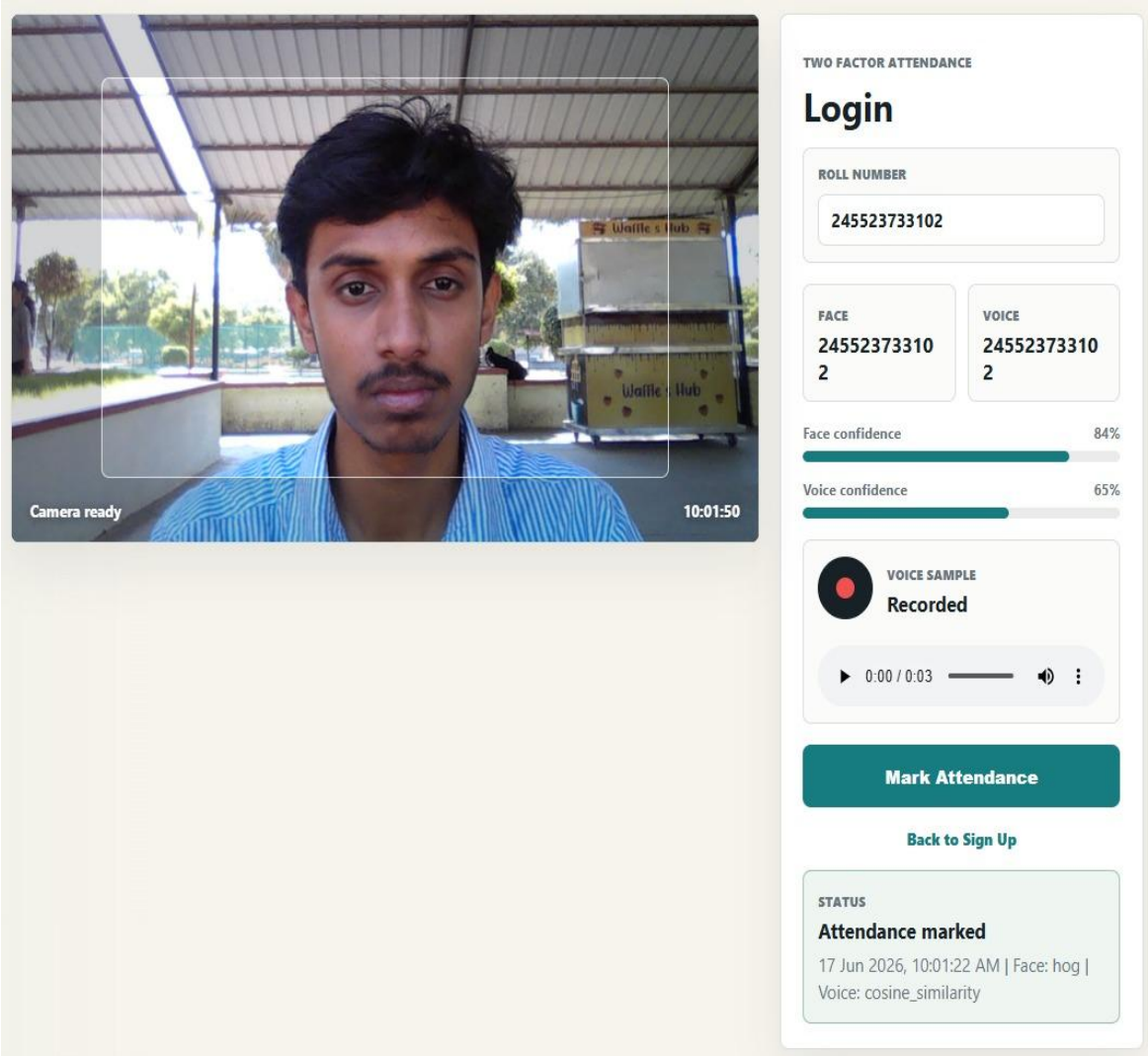


Fig 6.1.2 Student Login Interface

The login interface allows the student to enter the registered roll number and provide live face and voice samples. The system performs face recognition using HOG and dlib-based facial embeddings and verifies the voice sample using MFCC feature extraction and SVM classification. The similarity scores obtained from both biometric modalities are compared with the stored templates to validate the student's identity.

6.1.3 Attendance Marking and Verification Result

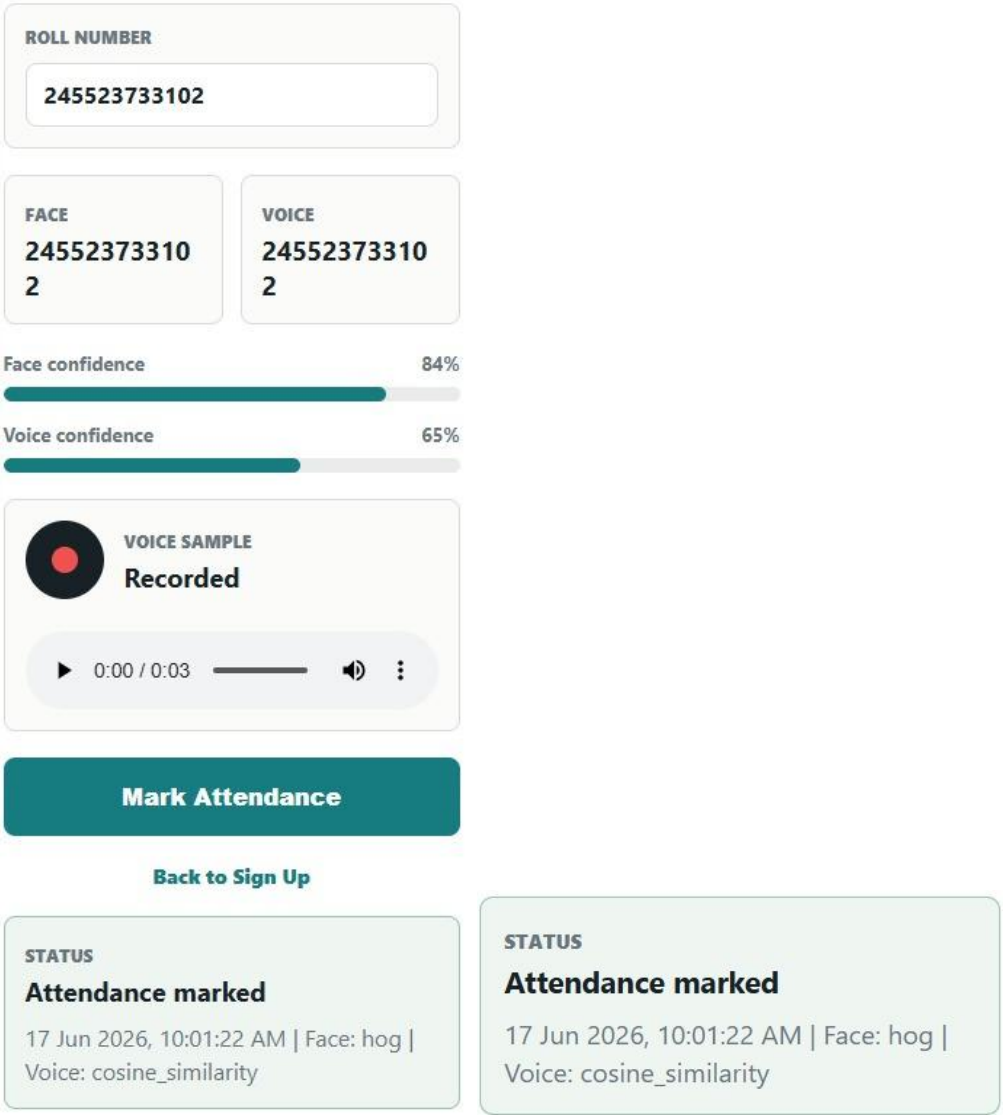


Fig 6.1.3 Attendance Marking Result

After successful face and voice authentication, the system grants access and automatically marks attendance. The attendance record contains the date, time, facial verification result, and voice verification result. This dual-factor authentication mechanism prevents proxy attendance and improves the reliability and accuracy of attendance management.

CHAPTER-7

7. CONCLUSION & FUTURE SCOPE

7.1 SUMMARIZE PROJECT FINDINGS

The project successfully develops a multimodal biometric attendance management system using face recognition and voice authentication technologies. The system utilizes facial embeddings generated through deep learning techniques and voice features extracted using MFCC to accurately verify the identity of students. By combining face recognition and voice verification, the system provides a secure two-factor authentication mechanism for attendance marking.

The proposed system effectively addresses the limitations of traditional attendance methods by reducing manual effort, minimizing human errors, and preventing proxy attendance. The integration of HOG-based face detection, dlib facial landmark recognition, Euclidean distance matching, MFCC feature extraction, and SVM classification improves the reliability and accuracy of biometric verification.

The attendance records are automatically maintained in a CSV file, making the system simple, efficient, and easy to manage. The project demonstrates the practical application of computer vision, machine learning, and biometric authentication in educational institutions. Overall, the system provides an accurate, cost-effective, and user-friendly solution for attendance management and can be further enhanced with real-time monitoring, cloud integration, and advanced anti-spoofing techniques in the future.

7.2 FUTURE SCOPE

The proposed Multimodal Identification Verification System is currently useful for schools and colleges, but it can be expanded to offices, hospitals, industries, libraries, and examination centers. In the future, it can be connected with cloud databases such as MySQL, Firebase, or MongoDB to manage large user records efficiently and provide centralized access to attendance data.

IoT devices like smart cameras, RFID gateways, and biometric terminals can be added for automated and real-time attendance. The system can also be improved using advanced deep learning models to increase face and voice recognition accuracy in different conditions. Anti-spoofing methods such as liveness detection and voice replay detection can strengthen security and prevent fraudulent access.

A mobile application can be developed for remote access to attendance records and notifications, while GPS-based verification can support remote attendance monitoring. Real-time dashboards and automated reports can help administrators track attendance easily and make better decisions. The system can also be integrated with leave management, payroll, and student information systems. These upgrades will make the solution more scalable, secure, and suitable for large organizations.

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